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Calibration results
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Normalized Residuals
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Reprojection error (cam0):     mean 0.1797439144300142, median 0.14532027357930255, std: 0.1306942495945175
Reprojection error (cam1):     mean 0.17949804866576935, median 0.142848817245052, std: 0.13418087701310635
Gyroscope error (imu0):        mean 0.2548463600950168, median 0.24167534535656937, std: 0.11736704008778906
Accelerometer error (imu0):    mean 0.1816939223199337, median 0.1637742481779656, std: 0.1049637668159428

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.1797439144300142, median 0.14532027357930255, std: 0.1306942495945175
Reprojection error (cam1) [px]:     mean 0.17949804866576935, median 0.142848817245052, std: 0.13418087701310635
Gyroscope error (imu0) [rad/s]:     mean 0.0015897968055127072, median 0.001507632645315412, std:
0.0007321656284770198
Accelerometer error (imu0) [m/s^2]: mean 0.025348580052384797, median 0.022848560851409606, std:
0.014643761396975865

Transformation (cam0):
-----------------------
T_ci:  (imu0 to cam0):
[[ 0.02938083 -0.99956733  0.00138766  0.01926999]
[-0.02109669 -0.00200806 -0.99977542  0.03160491]
[ 0.99934563  0.02934495 -0.02114656 -0.09098038]
[ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.02938083 -0.02109669  0.99934563  0.09102144]
[-0.99956733 -0.00200806  0.02934495  0.02199493]
[ 0.00138766 -0.99977542 -0.02114656  0.02964715]
[ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.012650680250590367


Transformation (cam1):
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T_ci:  (imu0 to cam1):
[[ 0.02887221 -0.99958254  0.00106809 -0.03071919]
[-0.0232713  -0.00174042 -0.99972767  0.03178553]
[ 0.99931218  0.02883949 -0.02331184 -0.09067077]
[ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.02887221 -0.0232713   0.99931218  0.09223503]
[-0.99958254 -0.00174042  0.02883949 -0.02803614]
[ 0.00106809 -0.99972767 -0.02331184  0.02969598]
[ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.012698216605296648

Baselines:
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Baseline (cam0 to cam1):
[[ 0.99999982  0.00033026 -0.00050198 -0.05004528]
[-0.00033135  0.9999976  -0.00216635 -0.00001002]
[ 0.00050126  0.00216651  0.99999753  0.00023126]
[ 0.          0.          0.          1.        ]]
baseline norm:  0.050045814036265006 [m]


Gravity vector in target coords: [m/s^2]
[-0.19340961 -9.80461183 -0.0245454 ]


Calibration configuration
=========================

cam0
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```

Camera model: pinhole
Focal length: [387.26496788634023, 388.0178836701606]
Principal point: [320.53538193017073, 238.7961117373796]
Distortion model: radtan
Distortion coefficients: [-0.0048964025986568405, -0.0032256738628326904, 6.867803461190135e-05, 0.0003187277311499132]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

Camera model: pinhole
Focal length: [388.18861658777746, 388.9712538925426]
Principal point: [321.2136971603922, 238.95658293993972]
Distortion model: radtan
Distortion coefficients: [-0.002870215094047284, -0.00471202846643744, 5.440321557997829e-05, -9.458602566352592e-05]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

IMU configuration
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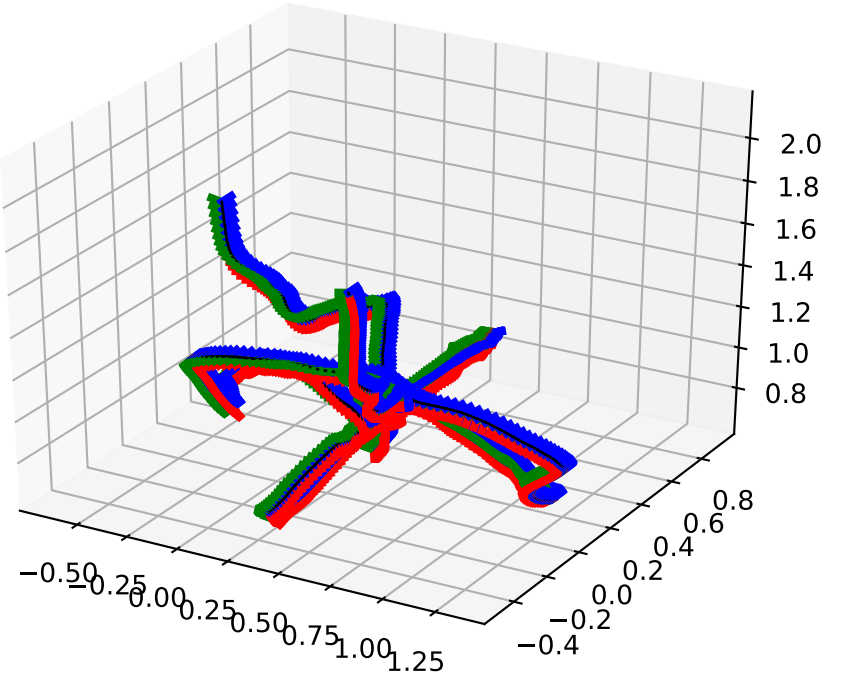
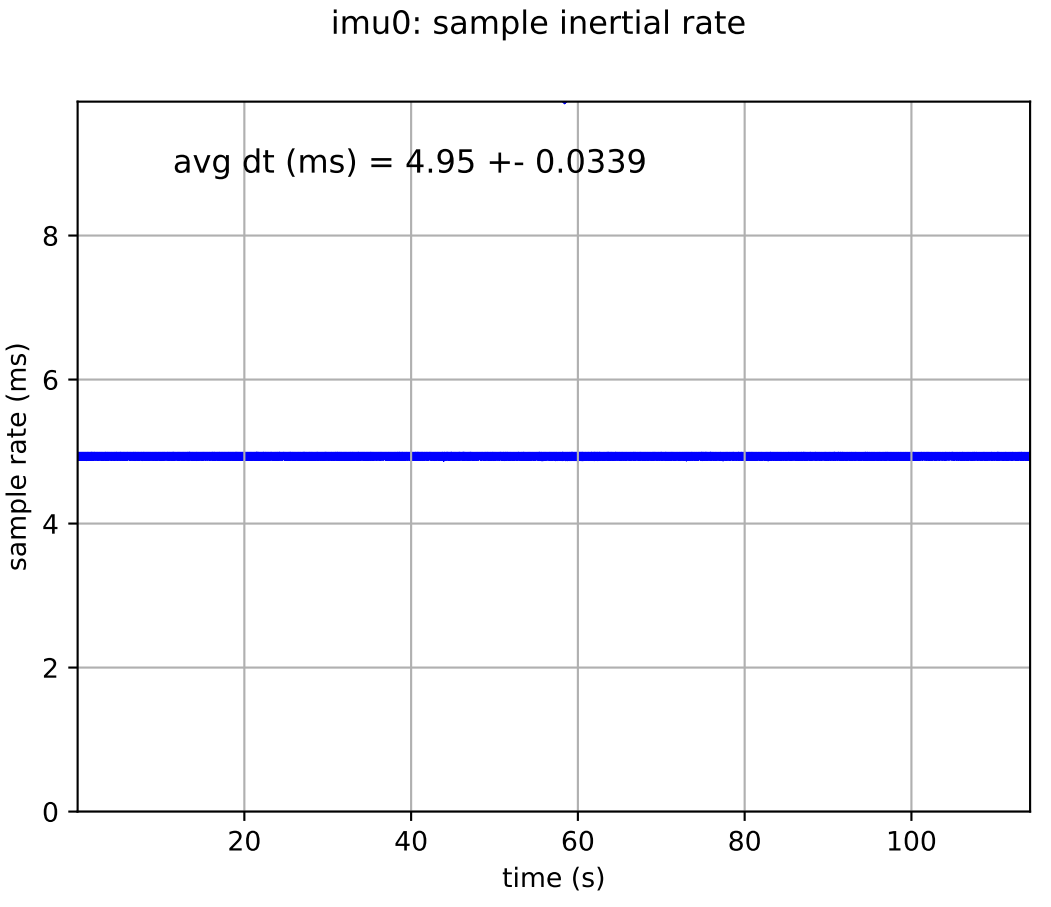
IMU0:

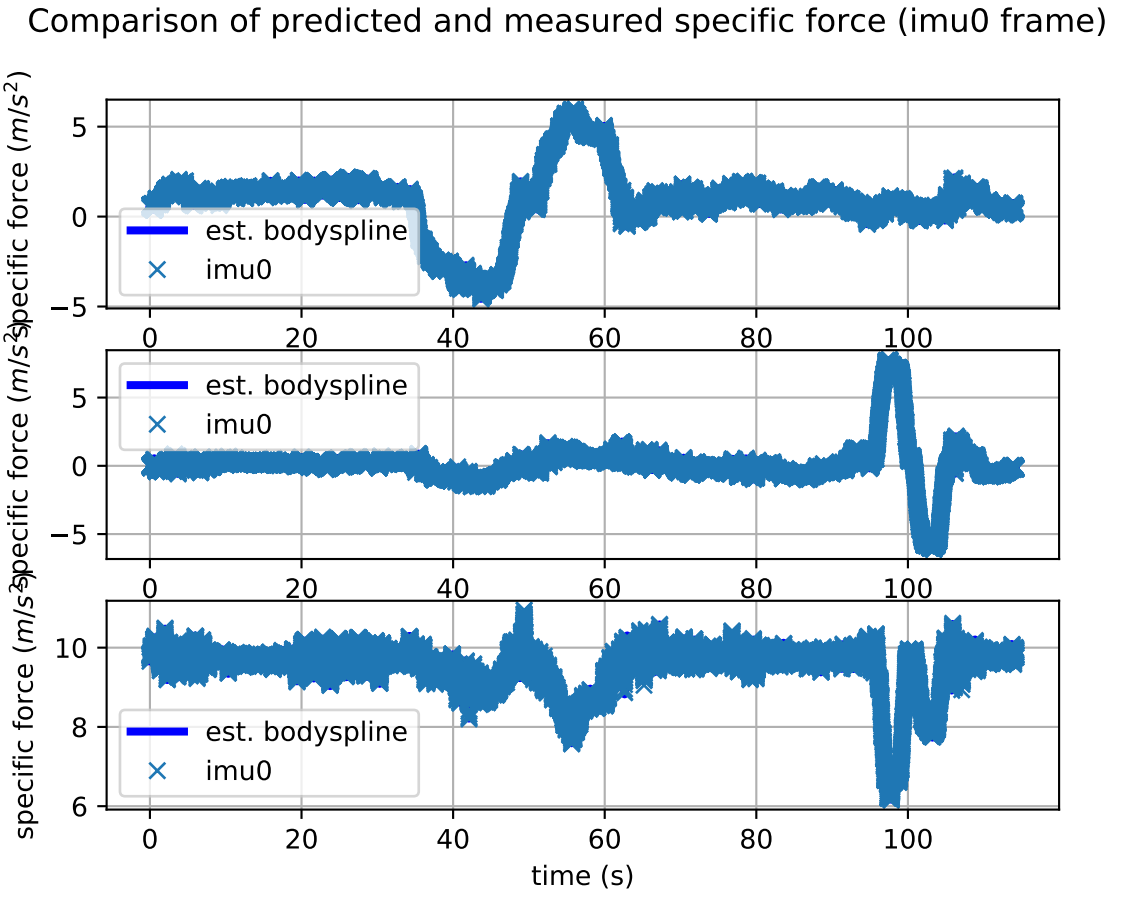
Model: calibrated
Update rate: 201.842746

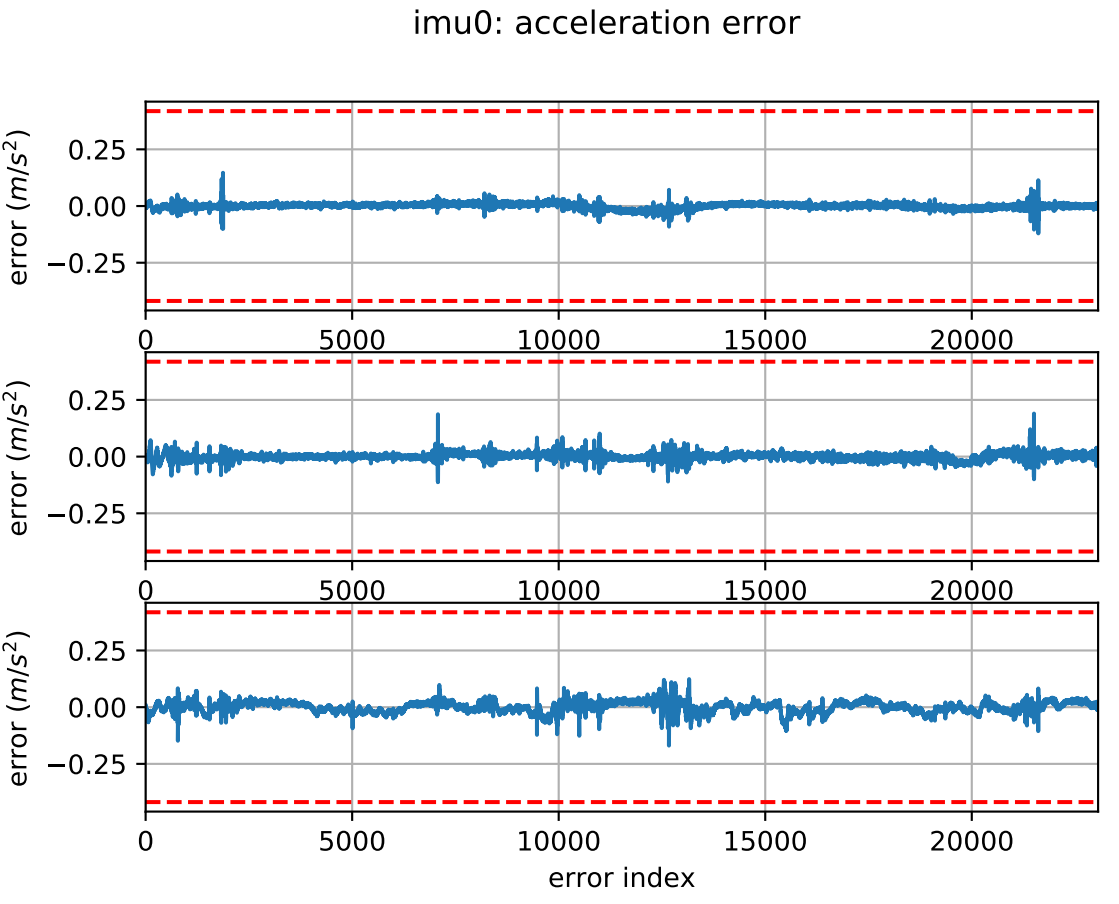
```
Accelerometer:
Noise density: 0.00981989090996197
Noise density (discrete): 0.13951253695624466
Random walk: 0.0005601293637217548
Gyroscope:
Noise density: 0.00043909309937990626
Noise density (discrete): 0.00623825588452575
Random walk: 1.6907369481278766e-05
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]
```

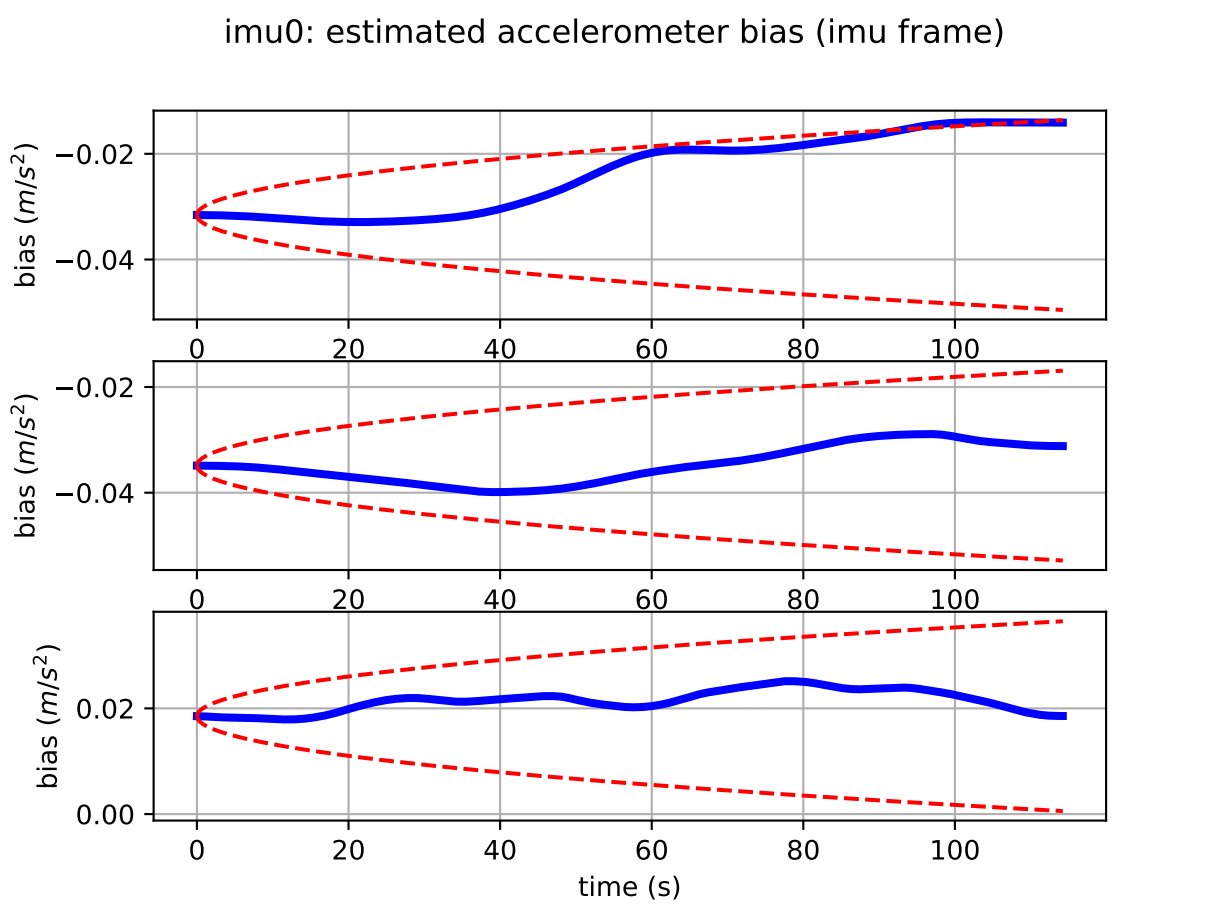
imu0: estimated poses



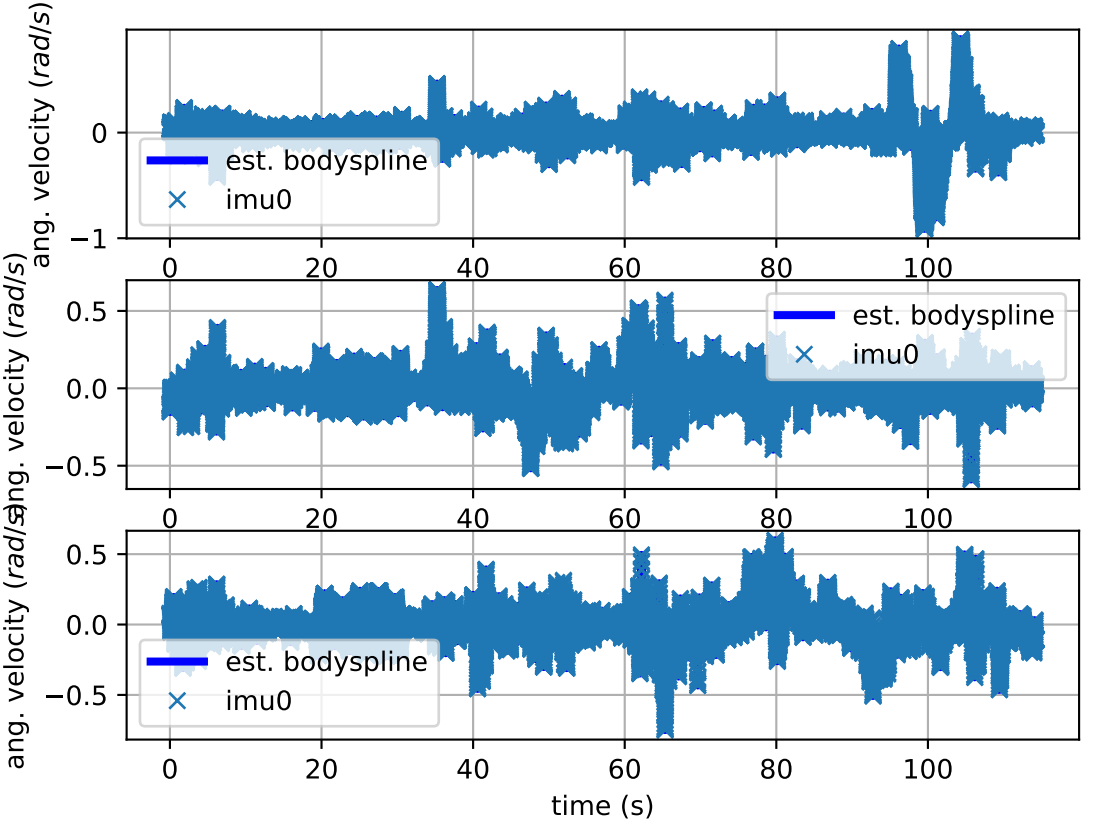

imu0: sample inertial rate
avg dt (ms) = 4.95 +- 0.0339
sample rate (ms)
0
2
4
6
8
20
40
60
80
100
time (s)

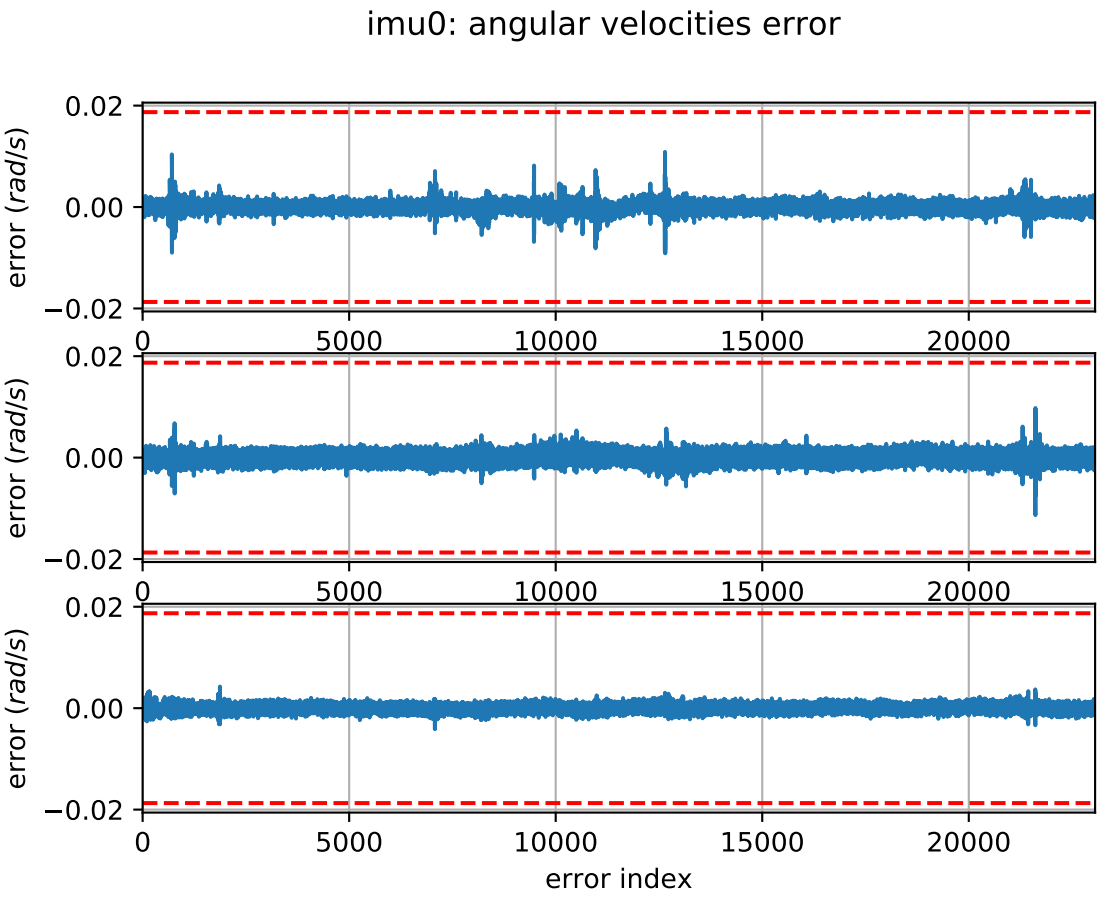

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
time (s)

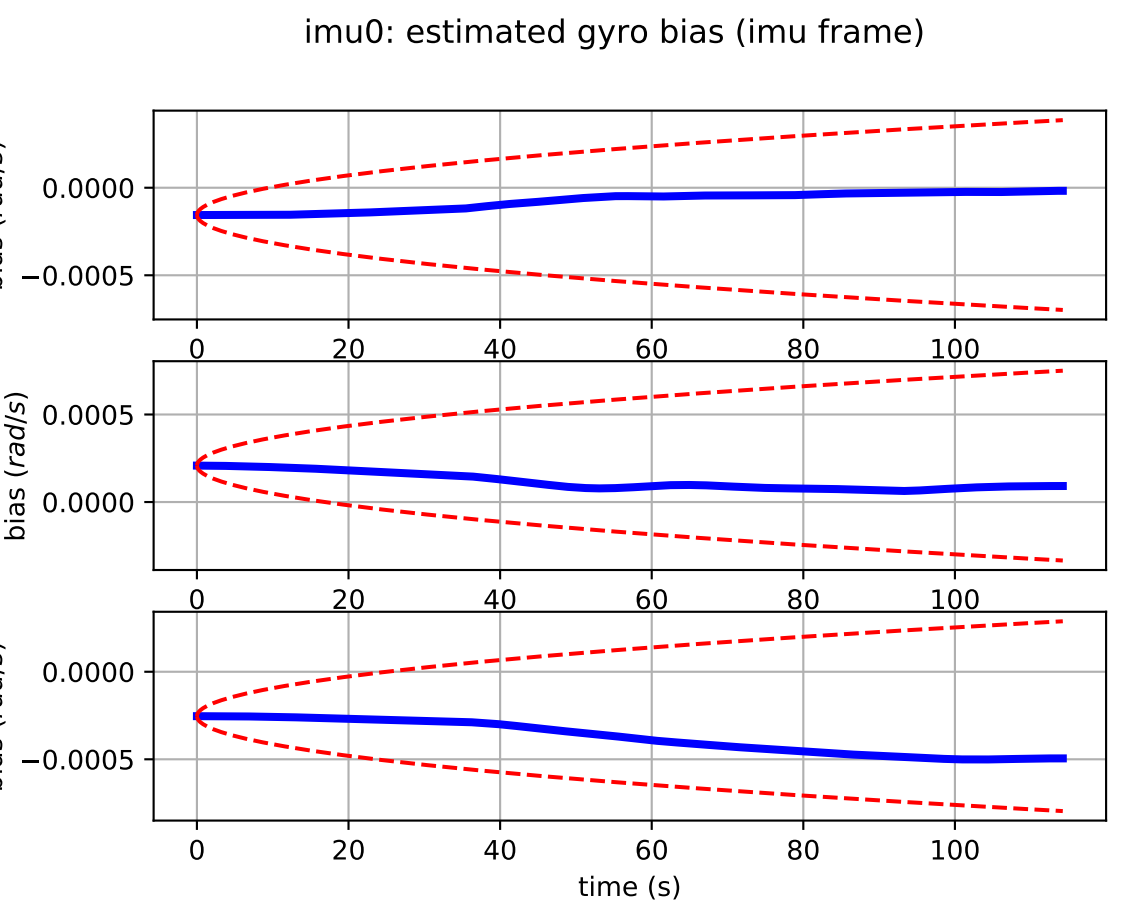

imu0: acceleration error
error (m/s^2)
0.25
0.00
−0.25
0
5000
10000
15000
20000
error index

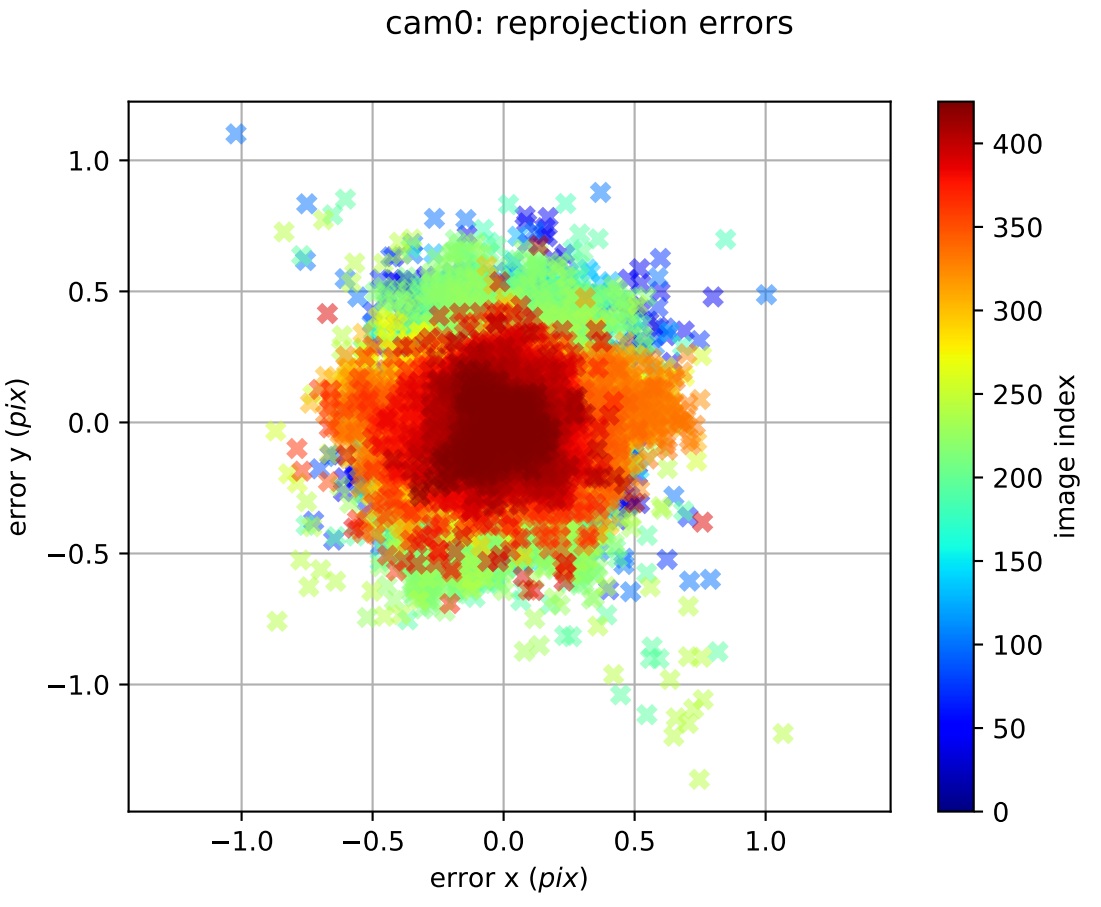

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
−0.02
−0.04
0
20
40
60
80
100
bias (m/s^2)
−0.02
−0.04
0
20
40
60
80
100
bias (m/s^2)
0.02
0.00
0
20
40
60
80
100
time (s)

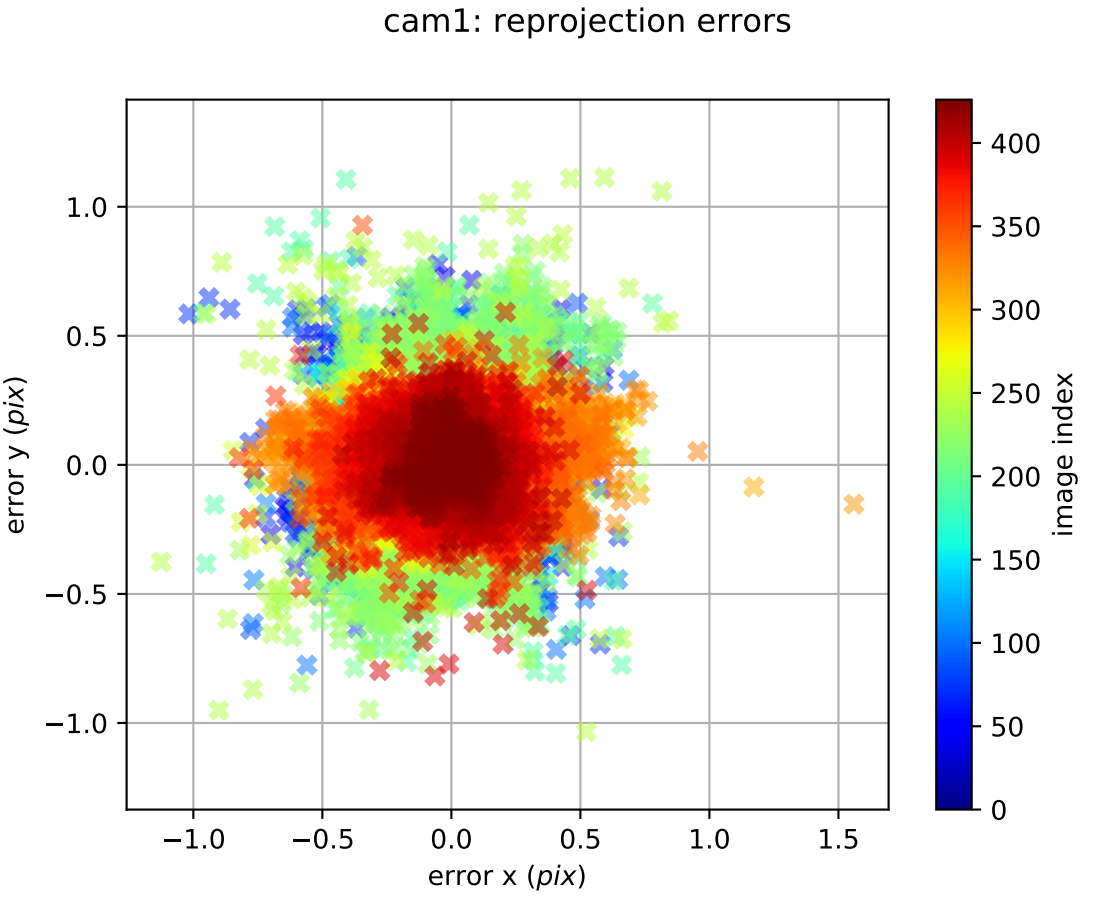
Comparison of predicted and measured angular velocities (body frame)




imu0: angular velocities error
error (rad/s)
0.02
0.00
−0.02
0
5000
10000
15000
20000
error index


imu0: estimated gyro bias (imu frame)
0.0000
−0.0005
0.0005
0.0000
0.0000
−0.0005
0
20
40
60
80
100
bias (rad/s)
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
400
350
300
250
200
150
100
50
0


cam1: reprojection errors
error y (pix)
error x (pix)
image index
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
400
350
300
250
200
150
100
50
0